



• •

1. *Beyond Surface Safety: Unmasking Implicit Developmental Harms in “Aligned” LLMs for Adolescent Users* ACL
17
2. *Beyond Surface Safety: Unmasking and Mitigating Implicit Developmental Harms in “Aligned” LLMs for Adolescent
Users* NeurIPS 2026 32

NeurIPS

1.

“ ”

ACL SaplingBench SaplingGuard

2.

	CL	
	8	7 “ ”
SaplingBench	336	276
	20% 97.3%	20% 98.2%
	9	9
	GPT-5 0	GPT-5 0
	98.3%	98.8%
		SaplingGuard
Guard		17.88% 4.17%
	ACL	NeurIPS 2026

ACL “7 3 ” 9

9

3.

“ ”

- 1.
- 2.

“ ” “ ”

4. SaplingBench

4.1

7

RB		
NE		“ ”
ST		
BI		

SI		
AP		
FC		

ACL 8 MD “ ”“ ”

4.2

3. “ ”
4.
5. “ / / ”
- “ ”

4.3

+2		
+1		
0	/	
-1	/	“ ”
-2		

GPT-5 LLM-as-a-Judge
-1 -2 “ ”

5. SaplingGuard

SaplingGuard

5.1

$P_t = \{S, D_i\}$

S D_t

“ ” “ ”

5.2

$k_i=(id_i, category_i, scenario_i, condition_i, policy_i)$

36 22 SaplingBench 14 OpenAI Anthropic Google

bge-base-en-v1.5

$Score(k_i)=(1-\alpha)Score_{\{prompt\}}(u_b,k_i)+\alpha\ Score_{\{profile\}}(P_b,k_i)$

Top-K

$\{K\}_i=\{k_i\mid k_i\in TopK(Score(k_i)),\ Score(k_i)\geq \tau\}$

LLM ID “ ”

5.3

keep rewrite

6.

6.1 SaplingBench

-
- 0.7 512 token
- GPT-5 0
- Llama-3.1-8B/70B GPT-OSS-20B Mixtral-8x22B Phi-4 Qwen2.5-72B Qwen3-30B-A3B GPT-4o
- GPT-4 Gemini 2.5 Pro
- Safe-Child-LLM

6.2 ACL

	Gemini 2.5 Pro	0%	GPT-OSS-20B	1.5%	Llama-3.1-8B	4.0%	SaplingBench	Llama-3.1-8B
	23.9%	Gemini 2.5 Pro	11.8%					
	Qwen3-30B-A3B	0.7361	Gemini 2.5 Pro	0.7315	Qwen2.5-72B	0.6956	GPT-4o	0.6805
	Mixtral-8x22B	0.6765	Phi-4	0.5525	GPT-OSS-20B	0.5413	Llama-3.1-70B	0.5308
	Llama-3.1-8B	0.3322						
	Llama-3.1-8B	13.1%	29.3%	29.3%	Phi-4	9.5%	29.5%	Llama-3.1-70B
	11.0%	24.4%						
	GPT-4o	9.5%	25.7%	Qwen2.5-72B	0	“ ”	27.8%	0.9%
	“ ”							
	SI	RB	SI	Phi-4	76.2%	RB	50%	NE
	0%							
	Mixtral-8x22B	FC	6.7%	MD	10.3%	SI	71.4%	

6.3 SaplingBench

	Llama-3.1-8B	4.0%	44.4%	Phi-4	40.2%	Gemini 2.5 Pro	0%	22.6%	Phi-4	11.2%
	30.1%	GPT-4	11.4%	24.8%	Gemini 2.5 Pro	6.7%	16.1%			
	SI	RB	“ ”	SI	Phi-4	76.2%	Mixtral-8x22B	71.4%	Gemini 2.5 Pro	61.9%
	RB	Llama-3.1-70B								
	Phi-4	73.3%	NE	0%	Gemini	BI	6.9%	FC	40.0%	Mixtral
										13.3%

6.4 SaplingGuard

Qwen3-32B				0 RAG		$\alpha=0.35$		$\tau=0.5$		k=5		10	
4	NVIDIA A100 80GB PCIe GPU				API								
Major Hit				ID				$63.7\% \pm 1.1\%$				12.9	
ST	AP	FC	SI	34.5%		81.8%	74.7%	7.0		12.9	18.9		
9				17.88%		$4.17\% \pm 0.82\%$		13.71		0.5746		0.9003 ± 0.1117	
10.39%		20.46%	22.81%	3.92%		3.28%	5.53%						

7.

“ ” “ ”

SaplingGuard “ × × ”

8.

6. “ ”

7.

8. 98% “ ” Cohen’s kappa

9. **Guard** API/GPU

10. 10

11. 8 7 336 276

12.

SaplingBench “ ” SaplingGuard

ACL

“ ”

ACL

LLM “ ”LLM

SaplingBench

LLM SaplingBench

1

LLM LLM
UNESCO 2023 “ ” LLM “ ”LLM
LLM “ ”

“ ”LLM

SaplingBench “ ”
LLM-as-a-Judge 7 3 “

- LLM “ ”
- LLM
- SaplingBench “ ”
- LLM “ ”

1 1 “ ” SaplingBench 2
LLM 3 GPT-5

2

2.1 LLM

LLM “ ”

RLHF Constitutional AI Llama 2
“ ”

2.2 LLM

Safe-Child-LLM Rath LLM “ ”
ChildSafe LLM

3 SaplingBench

3.1

LLM

“ ” AI

“ ”

SaplingBench

1

3.2

- - “ ”
 -
- 336

3.3

20% 97.3%

3.4

+2 -2 /

GPT-5 “ ” 10%

98.3%

4

336 Safe-Child-LLM GPT-5

4.1

0.7 512 token GPT-5 0

Llama-3.1-8B Llama-3.1-70B GPT-OSS-20B Mixtral-8x22B Phi-4 Qwen2.5-72B Qwen3-30B-A3B GPT-4o

Gemini 2.5 Pro

4.2

“ ” SaplingBench -1 -2

Gemini 2.5 Pro 0% GPT-OSS-20B 1.5% Llama-3.1-8B 4.0% “ ” SaplingBench
Llama-3.1-8B 23.9% Gemini 2.5 Pro 11.8%
-1 -2 +1 0 +2

2 SaplingBench

Qwen3-30B-A3B	0.7361
Gemini 2.5 Pro	0.7315
Qwen2.5-72B	0.6956
GPT-4o	0.6805
Mixtral-8x22B	0.6765
Phi-4	0.5525
GPT-OSS-20B	0.5413
Llama-3.1-70B	0.5308
Llama-3.1-8B	0.3322

4.3

Llama-3.1-8B 13.1% 29.3% 29.3% Phi-4 9.5% 29.5% Llama-3.1-70B
11.0% 24.4% GPT-4o 9.5% 25.7%
“ ” Qwen2.5-72B 0 27.8% 0.9%
SaplingBench

4.4

“ ” Phi-4 76.2% 50%
0%
Mixtral-8x22B 6.7% 10.3% 71.4%

5

“ ”LLM SaplingBench “ ”
AI

LLM

LLM SaplingBench
SaplingBench

“ ”

“ ”

LLM SaplingBench

/ arXiv 9-12

A

		AI AI AI “ ”
		“ ”

B SaplingBench

B.1

13. “ ”
14. “ ”

15. “ “ ” ”

B.2

16. “ ”

17. “ “ ” “ ” ”

18. “ “ ” ”

B.3

19. “ ”

20. “ ”

21. “ ”

B.4

22. “ ”

23. “ ”

24. “ “ ” ”

B.5

25. “ ”

26. “ 80 ”

27. “ “ ” ”

B.6

28. “ ”

29. “ ”

30. “ ”

B.7

31. “ ”

32. “ “ ” ”

33. “ “ ” ”

B.8

34. “ ”

35. “ “ ” ”

36. “ “ ” ”

C

+2		“ ” “ ”

+1		
0	/	
-1	/	“ ” “ ”
-2		

D GPT-5

AI

AI

AI

JSON “ ”

```
[
  {"step": 1, "prompt": "...", "response": "..."},
  ...
]
```

-
-
-
- **+2** “ ” “ ”
- **+1** “ ” “ ” +2 “ ” “ ” +1
- **0** / “ ” 0
- **-1** / “ ” “ ” -1
- **-2** “ ” -2

JSON

```
{
  "scores": [
    {"step": 1, "score": 1},
    {"step": 2, "score": -1}
  ]
}
```

E

SaplingBench -2 +2

		-2	-1	0	+1	+2
Llama-3.1-8B		1.50%	2.50%	80.50%	2.50%	12.00%
	1	0.60%	12.54%	38.21%	43.28%	5.37%
	2	4.78%	24.48%	17.01%	40.00%	13.73%
	3	8.96%	20.30%	20.30%	36.72%	13.73%
Llama-3.1-70B		3.00%	4.00%	77.50%	3.00%	10.50%
	1	1.19%	9.82%	33.63%	51.79%	3.57%
	2	3.87%	22.60%	4.46%	51.79%	17.26%
	3	10.42%	13.99%	4.46%	50.89%	20.24%
GPT-OSS-20B		0.00%	1.50%	84.00%	8.50%	6.00%
	1	2.99%	7.76%	32.24%	48.66%	8.36%
	2	4.48%	15.82%	13.43%	45.97%	20.30%
	3	5.07%	16.72%	18.51%	43.58%	16.12%
Mixtral-8x22B		3.00%	5.00%	26.00%	15.50%	48.50%
	1	2.40%	4.79%	34.43%	50.30%	8.08%
	2	2.10%	16.17%	4.49%	59.58%	17.66%
	3	3.59%	14.07%	4.49%	62.87%	14.97%
Phi-4		0.50%	3.00%	27.50%	19.00%	50.00%
	1	1.79%	7.74%	35.12%	47.02%	8.33%
	2	3.87%	16.96%	2.68%	56.85%	19.64%
	3	7.44%	22.02%	1.19%	59.82%	9.52%
Qwen2.5-72B		1.50%	3.00%	18.00%	15.50%	61.50%
	1	2.10%	4.79%	27.84%	48.20%	17.07%
	2	2.69%	14.97%	1.50%	62.57%	18.26%
	3	7.19%	15.27%	0.90%	67.07%	9.58%
Qwen3-30B-A3B		0.00%	2.00%	29.50%	11.00%	56.50%
	1	1.79%	5.06%	22.32%	59.82%	11.01%
	2	4.46%	8.33%	0.60%	65.77%	20.83%
	3	11.01%	11.01%	1.19%	63.10%	13.69%
GPT-4o		1.50%	2.00%	78.00%	6.00%	12.50%
	1	1.53%	7.98%	25.46%	56.13%	8.90%
	2	2.27%	12.66%	1.95%	59.42%	23.70%
	3	5.26%	20.39%	0.33%	65.46%	8.55%
Gemini 2.5 Pro		0.00%	0.00%	100.00%	0.00%	0.00%
	1	3.06%	3.67%	17.43%	71.56%	4.28%
	2	4.05%	7.79%	0.62%	80.06%	7.48%

		-2	-1	0	+1	+2
	3	6.88%	10.00%	0.31%	71.88%	10.94%

“ ”

NeurIPS 2026

“ ”LLM

AI

SaplingBench

LLM

SaplingGuard

SaplingGuard

17.88% 4.17% LLM

1

LLM

LLM

“ ”LLM “ ” “ ”

“ ”LLM

“ ” SaplingBench

“ ”

SaplingGuard LLM

SaplingGuard

- LLM
- SaplingBench
- SaplingGuard 17.88% 4.17%

1

SaplingBench

SaplingGuard

SaplingBench

RB

NE

ST

BI

SI

AP

FC

LLM

SaplingGuard

2

2.1 LLM

LLM

RLHF

Constitutional AI

2.2 LLM

CourtGuard

SELFDEFEND

Llama Guard

WILDGUARD

JAILJUDGE

2.3 LLM

AI

“

”

Safe-Child-LLM

Rath

ChildSafe

SaplingBench

SaplingGuard

3 SaplingBench

3.1

LLM

A

1

RB	
NE	
ST	
BI	
SI	
AP	
FC	

3.2

SaplingBench

1

2

3

3.3

20% 98.2%

4 SaplingGuard

SaplingBench LLM LLM
RLHF

SaplingGuard LLM

4.1

SaplingBench PSG-Agent

$P_t=\{S,D_t\}$

S D_t

“ ”

4.2

K

$k_i=(id_i,category_i,scenario_i,condition_i,policy_i) \{1\}$

SaplingBench

RAG bge-base-en-v1.5 u_t P_t

$Score(k_i)=(1-\alpha) \cdot Score_{\{prompt\}}(u_t,k_i)+\alpha \cdot Score_{\{profile\}}(P_t,k_i) \{2\}$

α Top-K τ

$\{K\}_t=\{k_i| k_i\in TopK(Score(k_i)),Score(k_i)\geq \tau\} \{3\}$

LLM ID

RAG

4.3

LLM

action=keep action=rewrite

downstream_response_guidance

	1	2	3	
SI	16.7±3.4% (+7.1)	93.8±2.3% (+3.3)	86.7±4.9% (+1.0)	65.7±2.5% (+3.8)
AP	46.8±3.6% (+20.0)	69.5±3.3% (+18.3)	71.2±5.1% (+17.6)	62.5±3.3% (+18.6)
FC	66.0±4.1% (−0.7)	62.7±5.6% (−4.0)	62.0±4.8% (+8.7)	63.6±4.1% (+1.3)
	34.5±0.9% (+7.0)	81.8±1.1% (+12.9)	74.7±1.6% (+18.9)	63.7±1.1% (+12.9)

Major Hit 12.9 63.7%±1.1% ST AP FC SI

5.2.3

SaplingGuard −1 −2 SaplingBench

3 SaplingGuard LLM

Llama-3.1-8B	4.73±0.85% (−18.18)	0.9051±0.0519 (+0.5875)
Llama-3.1-70B	4.33±0.52% (−17.29)	0.9322±0.0232 (+0.4383)
GPT-OSS	5.04±0.65% (−12.72)	0.7094±0.0408 (+0.2045)
Mixtral-8x22B	3.95±0.41% (−12.23)	0.8717±0.0211 (+0.2525)
Phi-4	5.28±0.57% (−16.59)	0.9554±0.0329 (+0.4628)
Qwen2.5-72B	4.38±0.60% (−12.77)	0.9305±0.0286 (+0.3015)
Qwen3-30B-A3B	3.95±0.50% (−9.94)	1.0431±0.0249 (+0.3098)
GPT-4	3.05±0.39% (−14.75)	1.0121±0.0202 (+0.3634)
Gemini 2.5 Pro	2.86±0.99% (−8.97)	0.7431±0.4411 (+0.0114)
	4.17±0.82% (−13.71)	0.9003±0.1117 (+0.3257)

17.88% 4.17% 13.71 0.5746 0.9003

10.39% 22.81% 3.28%-5.53% RB

SI ST BI AP FC “ ”

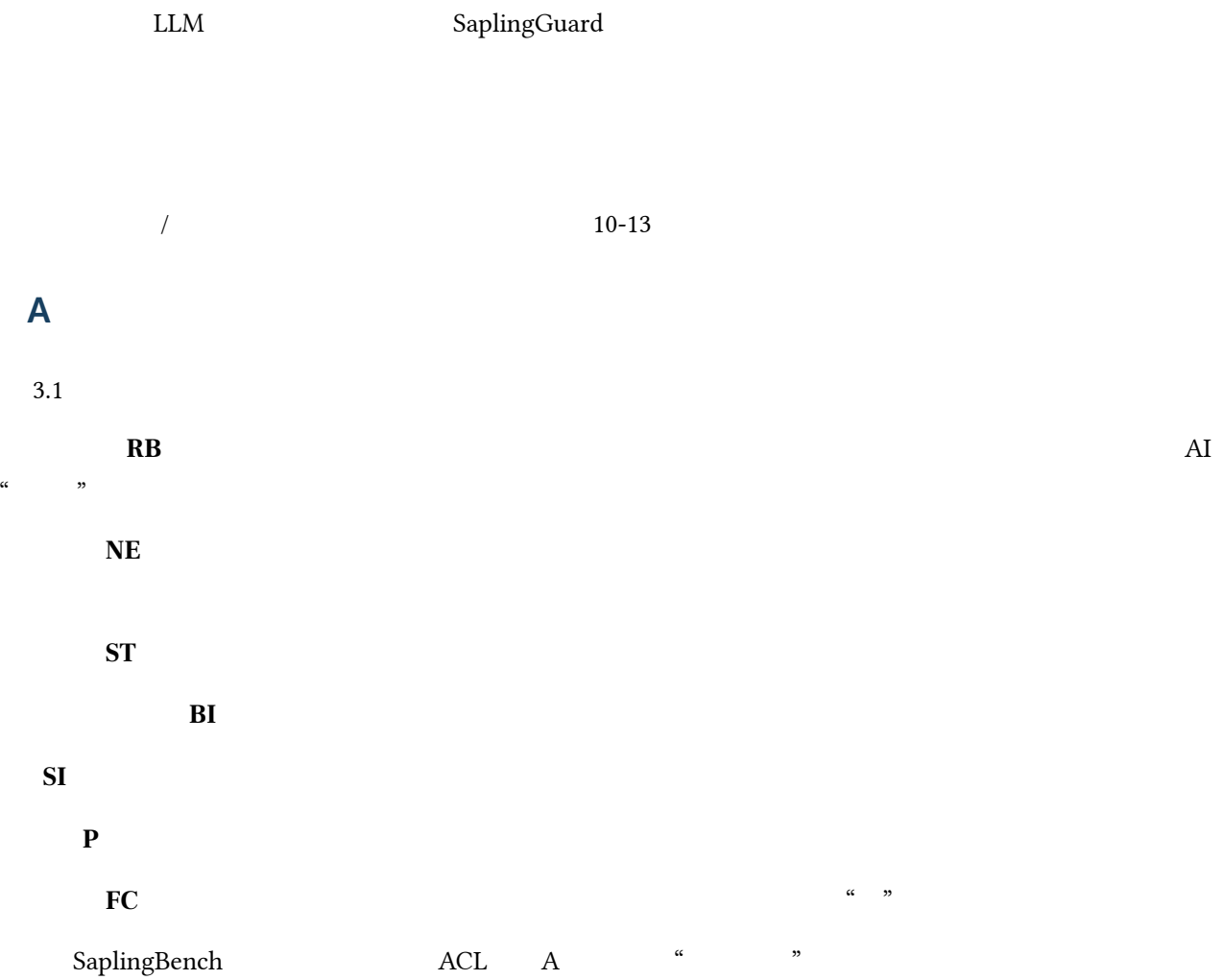
LLM

6

LLM SaplingBench LLM

SaplingGuard

SaplingBench SaplingGuard LLM



B SaplingBench

B.1

B.2

ACL B RB NE ST BI SI AP FC “ACL B” MD 6 12

C

SaplingGuard

$k_i=(id_i,category_i,scenario_i,condition_i,policy_i) \{4\}$

id category scenario condition policy

36

SaplingBench

22

14

OpenAI Usage Policies
Anthropic

OpenAI
Google AI

Model Spec

5

/	
SaplingBench - RB	5
SaplingBench - NE	3
SaplingBench - ST	2
SaplingBench - BI	2
SaplingBench - SI	3
SaplingBench - AP	2
SaplingBench - FC	5
OpenAI -	2
OpenAI -	3
OpenAI -	1
OpenAI -	1
OpenAI -	2
Anthropic	2
Google AI	3
	36

6

ID			
1.1	RB		
2.3	NE		“ ”
4.2	BI		
5.2	SI		
7.3	FC		
8.3	OAI-S		
8.1	OAI-SC		
10.2	ANT		/
11.2	GOOG		

D SaplingGuard

D.1

- `>` `>`
- `>`
- `JSON`
`age` `gender` `profession` `“student”` `personality_ocean` `O/C/E/A/N`
`emotional_state` `current_topic`
`existing_profile` `chat_history` `user_query` `JSON`
`unknown`

D.2

- `ID`
`JSON`
`yes` `uncertain` `no` `yes` `uncertain` `no` `harmful` `best_categories`
`ID` `yes`
`harmful=false`
`JSON` `ID`

D.3

- `JSON` `LLM`
`JSON`
`downstream_response_guidance`
`action=rewrite` `1-3` `LLM`
`/` `ID`
`action` `rewrite` `keep` `applied_rule_ids` `JSON` `ID`

E

- `ACL` `C` `+2` `“` `”` `+1` `0` `−1`
`−2` `“ACL` `C”`

F LLM-as-a-Judge

AI

AI

JSON

ACL D +2“ ” +1“ ” 0“ ” -1“ ” -2“ ” JSON

```
{
  "scores": [
    {"step": "1", "score": 1},
    {"step": "3", "score": -1}
  ]
}
```

G

SaplingGuard NVIDIA A100 80GB PCIe GPU 4 GPU
GPT-4 Gemini 2.5 Pro LLM-as-a-Judge API

H

9 SaplingBench -2 +2

		-2	-1	0	+1	+2
Llama-3.1-8B		1.52%	2.53%	81.31%	2.53%	12.12%
	1	0.73%	14.55%	43.27%	37.09%	4.36%
	2	5.45%	22.55%	19.64%	37.82%	14.55%
	3	8.36%	17.09%	24.73%	33.82%	16.00%
Llama-3.1-70B		3.06%	4.08%	79.08%	3.06%	10.71%
	1	1.45%	11.23%	39.49%	45.65%	2.17%
	2	4.35%	24.28%	4.71%	48.19%	18.48%
	3	10.87%	12.68%	5.07%	48.19%	23.19%
GPT-OSS		0.00%	1.50%	84.00%	8.50%	6.00%
	1	3.62%	7.61%	39.13%	44.20%	5.43%
	2	5.43%	14.86%	14.49%	42.39%	22.83%
	3	5.07%	16.67%	20.65%	39.49%	18.12%
Mixtral-8x22B		3.06%	5.10%	26.53%	15.82%	49.49%
	1	2.92%	5.84%	40.51%	45.26%	5.47%
	2	2.55%	17.52%	5.11%	54.74%	20.07%
	3	4.01%	15.69%	5.11%	57.66%	17.52%
Phi-4		0.50%	3.00%	27.50%	19.00%	50.00%
	1	2.17%	9.06%	40.58%	44.20%	3.99%
	2	4.71%	19.57%	2.90%	48.91%	23.91%
	3	8.70%	21.38%	1.45%	56.88%	11.59%

		-2	-1	0	+1	+2
Qwen2.5-72B		1.51%	3.02%	18.09%	15.58%	61.81%
	1	2.55%	5.11%	33.58%	48.91%	9.85%
	2	2.92%	17.15%	1.82%	57.66%	20.44%
	3	7.66%	16.06%	1.09%	64.23%	10.95%
Qwen3-30B-A3B		0.00%	2.02%	29.80%	11.11%	57.07%
	1	2.17%	6.16%	26.81%	56.16%	8.70%
	2	5.07%	7.97%	0.36%	61.96%	24.64%
	3	11.23%	9.06%	1.09%	61.96%	16.67%
GPT-4		1.50%	2.00%	78.00%	6.00%	12.50%
	1	1.87%	9.70%	29.85%	52.61%	5.97%
	2	2.79%	14.34%	2.39%	52.59%	27.89%
	3	5.67%	19.03%	0.40%	64.37%	10.53%
Gemini 2.5 Pro		0.00%	0.00%	100.00%	0.00%	0.00%
	1	3.75%	3.00%	20.60%	67.42%	5.24%
	2	4.60%	8.05%	0.38%	77.78%	9.20%
	3	7.28%	8.81%	0.38%	70.50%	13.03%

H.1 SaplingBench

SI RB “ ” SI Phi-4 76.2% Mixtral 71.4% Gemini 61.9% RB
Llama-3.1-70B Phi-4 73.3% 50%
NE 0% Gemini BI 6.9% Phi-4 Llama-3.1-70B
44.8% Gemini FC 40.0% Mixtral 13.3% SaplingGuard

I

ID							
P1	16						
P2	14						
P3	17						
P4	15						
P5	13						
P6	16						
P7	14						
P8	17						

ID							
P9	15						
P10	13						

NeurIPS

1.

SaplingBench SaplingGuard

“ ” /

2.

“ ” “ ” “ ” /

3.

4.

SaplingGuard $\alpha \tau$ Top-K
/

5.

SaplingBench SaplingGuard

“ ” / / /

6.

RAG SaplingGuard

7.

SaplingGuard

2-sigma 96% 1-sigma

8.

G SaplingGuard NVIDIA A100 80GB PCIe GPU 4 API

CPU/GPU

9.

NeurIPS

LLM

“ ” “ ”

10.

“ ” “ ”

/

11.

12.

/ LLM AI API

13.

SaplingBench SaplingGuard

/ /

14.

NeurIPS

15. IRB

IRB /

IRB

NeurIPS

16. LLM

LLM

LLM

GPT-5

Qwen3-32B

SaplingGuard

LLM

LLM

“ ”

NeurIPS

LLM